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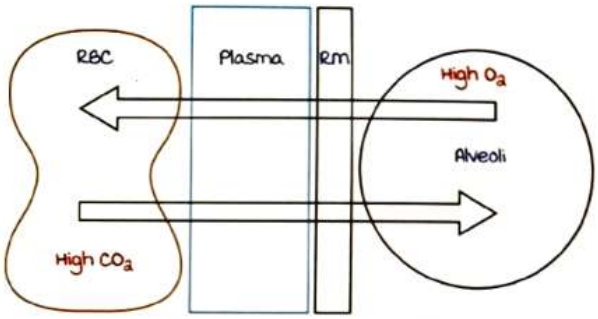
DIFFUSION AND TRANSPORT OF GASES

----- Active space -----

Diffusion of Gases

00:01:30

Transit time of blood in alveolar capillaries : 0.75 sec.
Therefore diffusion should take place in ≤ 0.75 sec.
Simple diffusion : Doesn't require ATP (Passive process).



Respiratory membrane (Rm)

Factors affecting Diffusion (Fick's law) :

Directly proportional
- Concentration gradient (High → low).
- Surface area of respiratory membrane.
- Lipid solubility of gas.

Inversely proportional
- Thickness respiratory membrane.
- Size of particle.

Applied aspect :

Emphysema :

α1 antitrypsin deficiency
↓
↑ trypsin activity (Protease)
↓
↑ destruction of Rm
↓
↓ membrane surface area

Pulmonary fibrosis :

Fibrosis
↓
↑ thickness of Rm

↓ diffusion of gases



Distented alveoli

DLCO :

- Test to assess diffusion capacity of lung using carbon monoxide (CO).
- CO : ↑ affinity to Hb (210 x O_2 affinity).
- Normal value : 25 ml/min/mmHg.
- Conditions affecting DLCO :

↑ DLCO

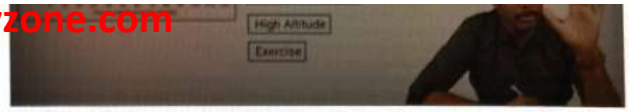
D/t ↑ blood flow
1. Exercise (physiological).
2. Polycythemia.
3. Pulmonary hemorrhage.

↓ DLCO

1. Anemia (d/t ↓ blood flow).
2. Emphysema (↓ area of Rm).
3. Pulmonary fibrosis (↑ thickness of Rm).

Note : Diffusion of CO_2 (20 times solubility of O_2) > O_2

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Transport of Gases

00:17:30

FORM OF TRANSPORT

O_a :

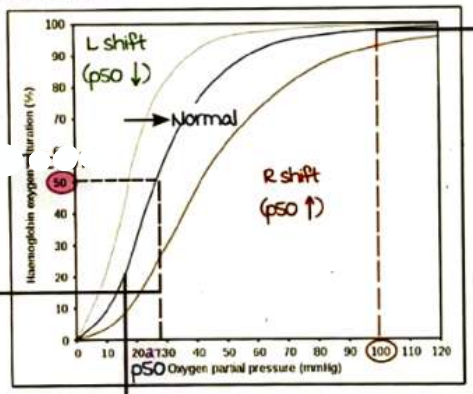
- **Plasma** (Dissolved form : 3 %)
  - Carries 3 % of O_a.
  - Determines pO_a.
- **Hemoglobin** (Combined form : 97 %)
  - Carries 97 % of O_a.
  - 1g Hb → 1.34 ml O_a/100 ml of blood.
  - ↑ O_a transport by 70 times.

CO_a :

- **Bicarbonate** (70 %) — Major form of transport
- **Hemoglobin** (30 %) — CO_a + Hb → carbamino Hb

O_a DISSOCIATION CURVES

O_a–Hb dissociation curve :



Plateau at pO_a : 100 mmHg as Hb is fully saturated

pSO : 50 % of Hb saturated at pO_a of 27 mmHg

Sigmoid curve

Sigmoid curve : 1 Hb molecule binds to 4 O_a molecules.

- 1st O_a — difficult binding → Hb (tight form)
- 2nd O_a — easy
- 3rd O_a — easy → Hb (relaxed form)
- 4th O_a — easy

Cooperative binding kinetics

Shifts :

	L shift (In lungs)	R shift (In tissues)
Effects	O_a — Binding / Loading → Hb Haldane effect: ↑ affinity of O2 to Hb → Loading of O_a → Unloading of CO_a	Release / Unloading → O_a ↓ affinity to Hb, Release of O_a (Bohr effect)
metabolic changes	↓ CO2 (Hypocarbia), ↓ H+ (Alkalosis)	↓ O_a (Hypoxia) → ↑ CO_a (Hypercarbia) → ↑ H^+ (Acidosis)

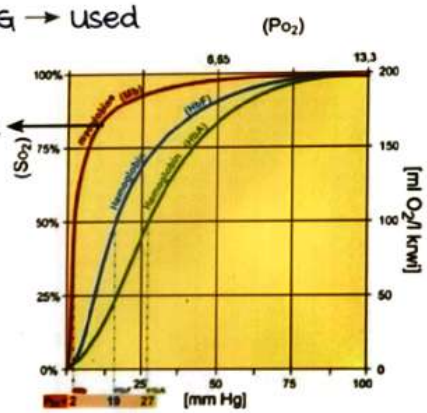
Feedback

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	L shift (In lungs)	R shift (In tissues)
Causes	1. Total Hb → ↓ affinity to 2,3 DPG. 2. CO poisoning. 3. Stored blood 4. Chronic hypoxia – glycolysis ↓ 2,3 DPG	1. Hypoxia → Anaerobic glycolysis ↑ 2, 3 diphosphoglycerate (2,3-DPG) Binds to Hb (β chain) O_2 release 2. Thyroid hormones → ↑ metabolism. 3. Exercise. 4. High altitude.

Note : citrate-phosphate-dextrose-adenine (CPDA) → Less fall in 2,3 DPG → used for blood storage.



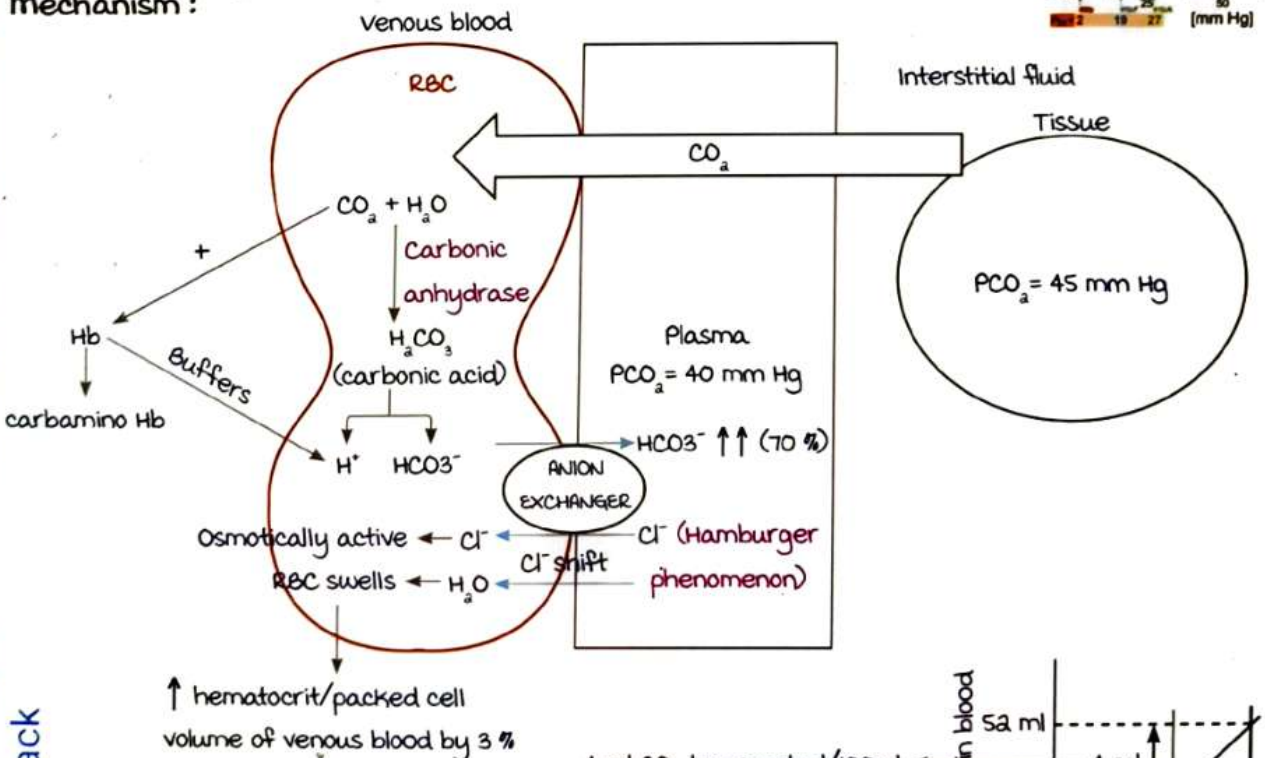
O_2 – myoglobin dissociation curve :

O_2 storage in muscles.

1 myoglobin molecule binds to 1 O_2 molecule (1 : 1) : No cooperativity.

CO_2 DISSOCIATION

mechanism :



CO_2 dissociation curve :

4 ml CO_2 transported/100ml of blood

